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Writing Effective Generative AI Prompts

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As a university student, please tell me how I can effectively write GenAI prompts.

Writing effective Generative AI (GenAI) prompts as a university student can help you get the most accurate and useful responses, whether for research, writing, coding, or creative tasks. Here's a structured approach to crafting high-quality AI prompts:

1. Be Clear and Specific

AI works best when given precise instructions. Instead of vague requests, clearly define what you need.

Example:

✗ Tell me about history.

✓ Summarize the causes and effects of World War II in under 200 words.

2. Provide Context

Adding context helps AI generate more relevant responses.

Example:

✗ Explain photosynthesis.

✓ Explain photosynthesis as if teaching a high school student, with simple examples.

Agenda

- Prompt Overview
- Frameworks
- Example Prompts
- Best Practices
- Tips
- Resources

“

Creating GenAI prompts is like painting
– you start with a blank canvas and see
where your creativity takes you!"

Microsoft Copilot

Common Terms

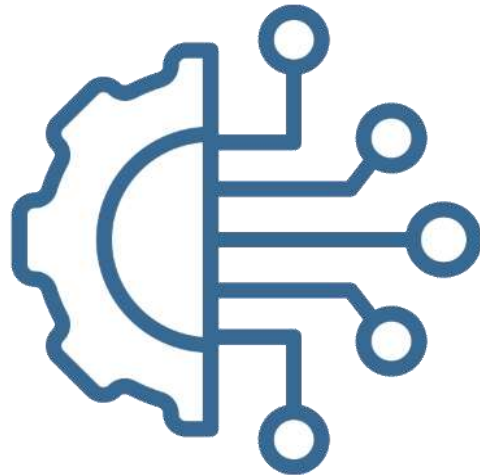
- **Generative Artificial Intelligence (GenAI):** AI that creates text, images, or other content based on patterns in data.
- **Large Language Model (LLM):** A type of AI trained on vast text data to generate human-like responses.
- **Temperature:** Adjusting this allows you to control how creative or conservative the AI's responses are. Higher values encourage creativity, while lower values ensure precision and reliability.
- **Top-p:** A sampling technique that restricts AI to the most probable choices for coherence. Higher values increase randomness & creativity but adds risk, while lower values ensure predictability and structure.
- **Tokens:** Units of text (words or subwords) processed by an AI model.
- **Reproducibility:** The ability to achieve the same AI-generated output given the same input and conditions.
- **Prompt (noun); prompting (verb):** A prompt is an input to AI, and prompting is the act of crafting inputs for desired responses.
- **Prompt Engineering:** The process of designing prompts to optimize AI outputs.
- **Framework (or Prompting Strategies/Techniques):** Structured approaches to writing prompts for better AI-generated responses.
- **Hallucinations:** When GenAI produces false, misleading, or nonsensical information that is not based on real data or facts.

How Prompts Work

Input = Prompt → GenAI Model → Output = Results



"Please explain the Super Bowl in simple terms."



The Super Bowl is the championship game of the National Football League (NFL) in the United States. Here's a simple breakdown:

1. **Teams:** It features the two best teams from the NFL season, one from the American Football Conference (AFC) and one from the National Football Conference (NFC).
2. **Timing:** The game is usually played on the first Sunday in February.
3. **Event:** It's not just a football game; it's a huge event with elaborate halftime shows, famous musical performances, and lots of commercials.
4. **Trophy:** The winning team gets the Vince Lombardi Trophy, named after a legendary NFL coach.

In short, the Super Bowl is a big, exciting football game that crowns the NFL champion and is celebrated with lots of entertainment and fanfare.

Are you planning to watch the Super Bowl this year?

Anatomy of a Prompt

As a

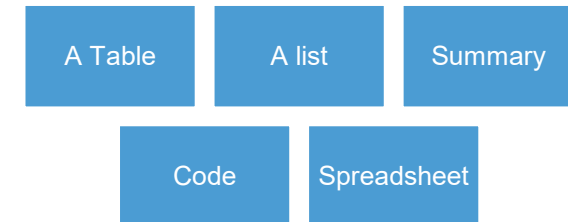
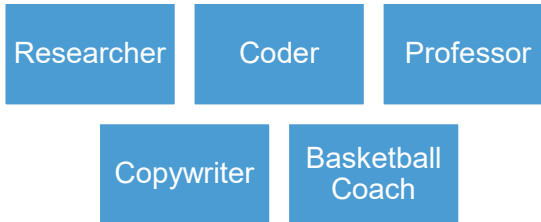
ROLE

perform

TASK

in

FORMAT



"You are a researcher analyzing AI ethics in healthcare."

"Explore how ethical frameworks guide AI in diagnostic tools and patient privacy, addressing key principles, challenges, and safeguards."

"Outline a well-structured academic essay that presents a clear thesis, supporting arguments, counterarguments, and a conclusion with policy recommendations."

Frameworks

CLEAR

Concise: Be brief and clear.

Logical: Create a logical flow and order of ideas.

Explicit: Specify output format, content or scope.

Adaptive: If at first you don't get what you want...

Reflective: Think about what worked (and didn't).

CO-STAR

Context: Provide background info.

Objective: Define the task clearly.

Style: Specify the desired writing style aligns the LLM response.

Tone: Ensure the response resonates with the required sentiment.

Audience: Tailor the LLM's response to a particular audience.

Response: Specify desired outputs.

TCREI

Task: Define the persona and the format of your output.

Format: What you want it to give you.

Context: Provide the reason for the ask, what you have tried.

References: Give a sample of what you are looking for.

Evaluate: Did you get what you wanted?

Iterate: Redirect.

Frameworks – Use Cases

CLEAR

Essay Writing & Research Papers: To get well-structured summaries, arguments, or explanations from AI.

Studying & Note-Taking: Asking AI for concise and logical explanations of complex topics

Programming Help: Writing clear, structured prompts to generate correct code or debug errors.

Proofreading & Editing: Refining AI-generated responses or academic writing with clear, adaptive feedback.

Exam Preparation: Asking for precise, logically structured answers to study questions

CO-STAR

Group Projects & Presentations: defining the style, tone, and audience to generate compelling content.

Email & Formal Writing: Drafting professional emails to professors, academic advisors, or internship applications

Creative Writing Assignments: Crafting AI-assisted stories, poetry, or speeches with a defined style and tone.

Debate & Persuasive Essays: Ensuring AI-generated strong arguments aligned with the intended audience.

Internship & Resume Building: Writing cover letters or resumes with an appropriate professional tone.

TCREI

Lab Reports & Research Projects: Reports follow a structured format with proper references

Coding Assignments & Debugging: Providing clear instructions and iterating to improve AI-generated code.

Data Analysis & Statistics: Generating insights or summarizing datasets.

Literature reviews: Summaries with cited references and evaluating for accuracy.

Essay Revision & Feedback: Iterating prompts to improve thesis statements, argument clarity, or citations.

Frameworks in Action: CLEAR

"What are the impacts of AI-based technologies on learning among students from diverse socioeconomic backgrounds in higher education?"

You are writing an essay about the impacts of AI-based technologies on learning among students from diverse socioeconomic backgrounds in higher education. Research a variety of AI educational tools including personalized learning platforms, virtual teaching assistants and intelligent tutoring systems. Consider the challenges of inequity in digital literacy, bias in AI systems, and threats to data privacy. Provide a detailed summary of current research incorporating recent published works and case studies. The summary should include examples of specific institutions, which tools are being used, and which learning outcomes are being measured. How do different institutional strategies and policy recommendations compare? What are their similarities and differences?

Concise
Logical
Explicit
Adaptive
Reflective

Frameworks in Action: CO-STAR

"How can information literacy empower individuals to recognize and resist misinformation amplified by AI-driven technologies?"

You host a podcast about information literacy. I am planning an episode that will discuss the ways misinformation is amplified by AI-driven technologies. Your goal is to educate listeners on the mechanisms through which AI spreads misinformation, and to empower them with strategies for developing digital resilience through information literacy. The podcast is straight forward and down-to-earth. It uses simple language with minimal jargon. The tone should be casual, friendly and upbeat. Your audience consists of adults, young and old, who have an interest in technology but are not necessarily tech-savvy. Provide five suggestions for an episode title along with an outline that includes cited sources and published research. It should discuss how to identify misinformation campaigns, how to verify sources before sharing information, and real-world strategies for developing critical thinking, education and awareness.

Context
Objective
Style
Tone
Audience
Response

Frameworks in Action: TCREI

"How should ethical frameworks guide the use of AI in healthcare decision-making, particularly in areas like diagnostic tools and patient privacy?"

Critically examine how ethical frameworks should guide AI decision-making in healthcare, particularly in diagnostic tools and patient privacy. Outline a well-structured academic research paper that presents a clear thesis, supporting arguments, counterarguments, and a conclusion with policy recommendations. AI is increasingly used in healthcare for diagnostics and patient data management, raising ethical concerns about bias, consent, privacy, and regulatory oversight. Ethical principles such as beneficence, autonomy, and justice must be considered to ensure AI's responsible implementation. Use at least three peer-reviewed sources, including ethical guidelines from organizations like the AMA, WHO, or HIPAA regulations. Cite recent studies on AI in healthcare. Discuss both the advantages and challenges of AI-driven decision-making. Analyze the effectiveness of current ethical guidelines in mitigating risks such as data privacy breaches, bias in AI algorithms, and the impact on human oversight in healthcare. Provide recommendations for improving AI ethics in healthcare. Suggest future areas of research or policy changes needed to ensure AI aligns with medical ethics and patient rights.

Task
Format
Context
References
Evaluate
Iterate



**Please Pick a Framework and Try
creating a Prompt**

Best Practices

Watch for Hallucinations.
Verify Accuracy.

Always follow Professor
and Workplace Guidelines.

ai.unc.edu

No Personal/Confidential Info.

Document GenAI Use.

Tips

Be Respectful. Use 'Please.'

Tell It To Think Before It Answers
and Cite Sources.

Ask For Clarification.

Iterate, Iterate, Iterate!

Use the Frameworks.

Provide Feedback On Your
Results.

Resources



The University of North Carolina at Chapel Hill



UNIVERSITY
LIBRARIES

REGISTER

library.unc.edu

News > Upcoming Events

heellife.unc.edu

Attend Events

Summer GenAI Series

Privacy, Property and Attribution in GenAI

- Tuesday, July 8, 11am-12pm ET

Gen AI Life Hacks

- Thursday, July 24, 2-3 pm ET

Prompts

Companies have collections of prompts to help users get started with GenAI quickly.



[Microsoft Copilot Prompt Gallery](#)



[Anthropic Prompt Library](#)



[Adobe Firefly Prompts](#)



[OpenAI Prompting Examples](#)



[Google AI Prompt Gallery](#)

Attribution Chart

Usage	Tool Used (e.g., ChatGPT-4)	How you edited the output, if at all	Conversation Link (If available)
Topic selection			
Brainstorming and idea generation			
Research			
Source valuation			
Outlining/planning			
Drafting			
Media creation			
Peer review			
Revising			
Polishing			
Other			

Courses

> LinkedIn:

- Create PowerPoint presentations from a Copilot prompt by Heather Severino
- Getting Hands-on with GPT 4 Tips and Tricks by Densys Linkov
- How to write an effective prompt for AI by David Birss

• Generative AI for Beginners:

https://learn.microsoft.com/en-us/collections/zpy7c8zmq6ky0z?WT.mc_id=academic-105485-koreyst

• The Art of ChatGPT Interactions: The SAGE Campus tutorial by Dr. Leo Lo of the University of New Mexico. Go to SAGE Research

Methods <https://guides.lib.unc.edu/go.php?c=23609079>.

- Click on **Campus** in the header to see the full list of courses, then select **The Art of ChatGPT Interactions_ course**.

> What Is the 5-Step Prompt Framework and How Does It Work: Google Prompting Essentials:

https://youtu.be/lp-Ft3Ex5_k?si=vNWYjFtQDTH22Ut6

• Struggling with writing a perfect ChatGPT prompt:

https://www.linkedin.com/posts/hamna-aslam-kahn_struggling-with-writing-a-perfect-chatgpt-activity-7218585149272166402-bT5r?utm_source=share&utm_medium=member_desktop

> Your guide to mastering All the different prompt frameworks: <https://juuzt.ai/knowledge-base/prompt-frameworks/>

Prompt Frameworks

- **A framework for enhancing information literacy through prompt engineering: The CLEAR framework: Concise, Logical, Explicit, Adaptive, and Reflective:**
<https://guides.lib.unc.edu/GenAI/prompts>
- **Generative Artificial Intelligence United State Military Academy Library: CO-STAR Framework:**
<https://library.westpoint.edu/GenAI/prompting>
- **Generative AI: Prompt Engineering Framework and Best Practice:**
<https://medium.com/@write2piri/generative-ai-prompt-engineering-framework-and-best-practices-3c4f170da6cd>
- **Your AI isn't broken – your prompts are:**
https://www.linkedin.com/posts/adam-bidd_your-ai-isnt-brokenyour-prompts-are-activity-7280628957270441986-Xzdo?utm_source=share&utm_medium=member_desktop
- **Struggling with writing a perfect ChatGPT prompt:** https://www.linkedin.com/posts/hamna-aslam-kahn_struggling-with-writing-a-perfect-chatgpt-activity-7218585149272166402-bT5r?utm_source=share&utm_medium=member_desktop
- **Your guide to mastering All the different prompt frameworks:** <https://juuzt.ai/knowledge-base/prompt-frameworks/>

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